Recursion categories and their coalgebraic representations

A catalogue and classification problems over sets of integers

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Abstract

We catalogue classical, arithmetical, analytical, admissible, syntactic, and operator examples of recursion categories, together with equivariant and existence constructions. The operator categories of Montagna and Sorbi give direct connections with effective descriptive set theory and Wadge reducibility. We organize coalgebraic representations and their comparisons in a 2-category, classify representations with constant coalgebra targets up to equivalence, and construct representations using binary observations of subsets of the integers. Countable recursion categories admit countable coalgebraic representations. Faithfulness requires retaining distinctions between states with equal observable behavior. The catalogue separates the underlying recursion category, its presentation, and the structure preserved by a representation. Pointed Moore coalgebras also present strictly causal operators on streams; we give their countable and computable realizations and relate their equality to behavioral equivalence.

1 The objects to be classified

A catalogue of recursion categories must specify both arrows and their equality. These determine which computations the category admits and which computations it identifies. A coalgebraic representation adds states, observations, and transitions; its comparison theory should retain the partial-map structure that supports universal computation.

1.1 Sets, sequences, and numerical codings

We work in ZFC and write $\omega = \{0, 1, 2, \ldots\}$ and $2 = \{0, 1\}$. Baire space is $\mathbb{B} = \omega^\omega$ and Cantor space is 2^ω, both with the product topology of their discrete coordinate spaces. A finite prescription of coordinates defines a basic clopen set. We identify a subset $A \subseteq \omega$ with its characteristic function $\chi_A \in 2^\omega$.

Fix the Cantor pairing bijection

$$\langle -, - \rangle : \omega^2 \longrightarrow \omega, \qquad \langle m, n \rangle = \frac{(m+n)(m+n+1)}{2} + n. \tag{1}$$

On the diagonal $m+n = s$, its values are the consecutive integers from $s(s+1)/2$ to $(s+1)(s+2)/2-1$. These intervals partition ω, proving bijectivity and giving a computable inverse. For each fixed arity, tuples are coded by iterating this pairing.

For a total function $f : \omega \to \omega$, define its coded graph $\Gamma_f \subseteq \omega$ and graph characteristic function $\chi_f : \omega \to 2$ by

$$\Gamma_f = \{\langle m, n \rangle : f(m) = n\}, \qquad \chi_f(\langle m, n \rangle) = \begin{cases} 1, & f(m) = n, \\ 0, & f(m) \neq n. \end{cases} \tag{2}$$

Thus $\chi_f = \chi_{\Gamma_f}$ is a point of Cantor space. The graph codes of all total functions form the subspace

$$\mathcal{G} = \{x \in 2^\omega : \text{for every } m \text{ there is exactly one } n \text{ with } x(\langle m, n\rangle) = 1\}.$$

This subspace is G_δ, meaning a countable intersection of open sets. Indeed, the condition of at most one 1 in each row is closed, and the condition of at least one is

$$\bigcap_{m\in\omega} \bigcup_{n\in\omega} \{x : x(\langle m, n\rangle) = 1\}.$$

Closed subsets of Cantor space are G_δ, so their intersection has the asserted form. The map $f \mapsto \chi_f$ is a homeomorphism $\mathbb{B} \to \mathcal{G}$: each graph coordinate tests one value of f, and the inverse image in $\mathcal{G}$ of the condition $f(m) = n$ is the clopen coordinate condition $x(\langle m, n\rangle) = 1$. Both maps are computable; to recover $f(m)$ from χ_f, search row m for its unique 1.

For $f, g \in \mathbb{B}$, interleaving is the function $(f \oplus g)(2n) = f(n)$ and $(f \oplus g)(2n+1) = g(n)$. Its inverse projections are $P_1 h(n) = h(2n)$ and $P_2 h(n) = h(2n+1)$. Write $f^+(n) = f(n+1)$ for the tail. To code subsets of $\mathbb{Z}$ by subsets of ω, use the bijection $\zeta : \mathbb{Z} \to \omega$ given by $\zeta(n) = 2n$ for $n \geq 0$ and $\zeta(n) = -2n - 1$ for $n < 0$.

1.2 Partial maps and recursion categories

A partial function $f : X \rightharpoonup Y$ is a function $D_f \to Y$ with $D_f \subseteq X$. Its source is X and its domain of definition is D_f. Its restriction $\overline{f} : X \rightharpoonup X$ is the identity on D_f. Composition gf applies f first; its domain is $\{x \in D_f : f(x) \in D_g\}$.

Abstractly, a restriction category assigns $\overline{f} : X \to X$ to each arrow $f : X \to Y$, subject to

$$f\overline{f} = f, \qquad \overline{f}\,\overline{g} = \overline{g}\,\overline{f}, \qquad \overline{g\overline{f}} = \overline{g}\,\overline{f}, \qquad \overline{g}f = f\overline{gf}.$$

The middle two equations concern arrows with the same source; the last concerns composable $f : X \to Y$ and $g : Y \to Z$. An arrow is *total* if $\overline{f} = 1_X$. A *restriction idempotent*, or *domain*, is an arrow $e : X \to X$ with $e = \overline{e}$. Domains are ordered by $e \leq d$ when $ed = e$. For parallel arrows, the restriction order is $f \leq g$ when $f = g\overline{f}$; for partial functions this means that g extends f.

We use the restriction-product formulation of a P-category [1, Proposition 4.5]. A binary near product $X \boxtimes Y$ has total projections p_X, p_Y and, for every $f : A \to X$ and $g : A \to Y$, a unique arrow $\langle f, g\rangle : A \to X \boxtimes Y$ satisfying

$$p_X\langle f, g\rangle = f\overline{g}, \qquad p_Y\langle f, g\rangle = g\overline{f}, \qquad \overline{\langle f, g\rangle} = \overline{f}\,\overline{g}.$$

Here angle brackets pair arrows; in (1) they pair natural numbers. Set $\Delta_X = \langle 1_X, 1_X\rangle$ and $f \boxtimes g = \langle fp_X, gp_Y\rangle$. The total arrows form a category $\mathrm{Tot}(\mathcal{C})$, in which these are binary categorical products. No nullary product is assumed.

A P-category is *prodominical* if it has zero arrows, absorbing under composition, and $f \boxtimes 0 = 0$ for every typed pair. These equations imply $\overline{0} = 0$. A nonzero object has $1_X \neq 0$; the category is nontrivial if it has such an object. An *isotype* is a nonempty category in which every two objects are isomorphic. A *Turing morphism* is an arrow $\tau : W \boxtimes X \to Y$ such that, for every object V and every $f : V \boxtimes X \to Y$, there is a total $g : V \to W$ with

$$f = \tau(g \boxtimes 1_X).$$

A *recursion category* is a prodominical P-category that is an isotype and has a Turing morphism [6, §9].

Di Paola–Heller's dominical convention additionally requires $f \boxtimes g = 0$ only when $f = 0$ or $g = 0$ and characterizes total arrows by $fh = 0 \Rightarrow h = 0$ for every composable h [2, Definitions 2.0–2.3]. The P-recursion categories of Di Paola–Montagna need not be dominical. We retain this distinction in the catalogue.

A range of $f : X \to Y$ is a domain $r : Y \to Y$ such that $rf = f$ and, for every parallel pair $u, v : Y \to Z$, the equality $uf = vf$ implies $ur = vr$. Coproducts below are categorical binary coproducts in the partial-map category. They are distributive if the canonical map $(A \boxtimes X) + (A \boxtimes Y) \to A \boxtimes (X + Y)$ is an isomorphism. A functor preserves near products or coproducts when the corresponding canonical comparison maps are isomorphisms.

Several properties distinguish the historical examples. A section of $f : X \to Y$ is an arrow $s : Y \to X$ with $fs = \overline{s}$ and $fsf = f$; *choice* means that every arrow has a section. An arrow $f : X \to Y$ is a partial monomorphism if $fu = fv$ implies $\overline{f}u = \overline{f}v$ for every parallel $u, v : Z \to X$. *Weak choice* means that every partial monomorphism is a partial isomorphism: it has $s : Y \to X$ with $sf = \overline{f}$ and $fs = \overline{s}$. An *atom* is a nonzero total arrow $a : X \to Y$ such that ea is either 0 or a for every domain e on Y. A *constant* is a total $c : X \to Y$ with $cu = cv$ for every pair of total arrows $u, v : Z \to X$. There are *enough atoms* (respectively, constants) if precomposition by all atoms (respectively, constants) with target Y separates arrows out of Y [2, §§6,8–9].

A nonzero domain e is *connected* if a disjoint decomposition $e = e_0 \vee e_1$ into domains forces one summand to be zero. Here $\vee$ is the domain join when the indicated join exists. The connected-domain separation condition says that $fh = gh$ for all connected domains h on the common source implies $f = g$. This is the condition called *locally connected* in the cited recursion-category literature [5, §4].

The catalogue is organized around the following data: its objects and arrows; equality; the near product and Turing morphism; domains and ranges; choice, coproducts, and connectedness; and a specified coalgebraic representation. The initial catalogue below includes the constructions of Di Paola–Heller, Montagna, Montagna–Sorbi, and Di Paola–Montagna.

2 The catalogue

2.1 Numerical and higher-recursion examples

The classical category $\mathcal{T}$. Di Paola–Heller's objects are countably infinite sets with recursive structures, drawn from a product-closed family; arrows are partial recursive maps in those structures, with extensional equality [2, Theorem 4.3, pp. 601–602]. Positive finite powers of ω give a concrete equivalent presentation. Pairing gives the near product, and a universal partial recursive function together with the parameter theorem gives a Turing morphism. The domains are recursively enumerable sets. Numerical search gives range sections.

Oracle relatives $\mathcal{T}_A$. For a fixed $A \subseteq \omega$, replace partial recursive maps by partial A-recursive maps on positive finite powers of ω. Equality remains extensional. The oracle universal machine and parameter theorem prove the recursion axioms by the same argument. These categories have ranges and sections obtained by dovetailing relative to A.

These concrete presentations detect Turing reducibility exactly:

$$\mathcal{T}_A \subseteq \mathcal{T}_B \quad \Longleftrightarrow \quad A \leq_T B. \tag{3}$$

The inclusion means identical objects and inclusion of the actual partial functions. One implication follows by simulating A-queries using B. Conversely, the characteristic function of A is an arrow of $\mathcal{T}_A$

and hence, under the inclusion, is B-recursive. Equation (3) concerns these numerical presentations; classification under arbitrary abstract equivalence is a further question.

The arithmetical hierarchy. For $n \geq 1$, Di Paola–Heller construct $\mathcal{T}(\Sigma^0_n)$, whose arrows have lightface Σ^0_n graphs in the chosen numerical structures. The domains are Σ^0_n sets, and the decidable domains are Δ^0_n sets. Their enumeration argument supplies a Turing morphism [2, §4.6, Example 4, p. 606]. For fixed numerical codings these are also the oracle categories $\mathcal{T}_{0^{(n-1)}}$: a functional graph is c.e. in the indicated jump exactly when its partial function is computable in that jump.

Analytical examples on integers. Di Paola–Heller construct $\mathcal{T}_\omega(\Pi^1_1)$ and state the corresponding construction of $\mathcal{T}_\omega(\Sigma^1_2)$ [2, §4.6, Example 5, pp. 606–607]. Arrows are partial numerical functions whose graphs lie in the indicated lightface pointclass. The construction of a universal functional relation uses single-valuedness results for those pointclasses. The subscript ω specifies that the free arguments are integers. This is a direct entry point for effective descriptive set theory: uniformization and universal relations supply the Turing morphism.

Admissible recursion. For admissible α, Di Paola–Heller describe $\mathcal{T}^\alpha$, using carriers that are copies of L_α with α-recursive structures and partial α-recursive maps [2, §4.6, Example 7, p. 608]. The admissible set and its coding belong to the catalogue entry. When L_α is countable, an external integer coding is available; the coding and the internal notion of α-recursiveness are separate data.

2.2 Valid syntactic categories

Montagna's S. The objects are generated from ω by binary products and disjoint sums. Arrows are program indices modulo a single PA-provable universally quantified equality of partial computations [3, pp. 105–108]. This is the case S_{PA} of the following construction.

Di Paola–Montagna's S_T. Let T be a consistent recursively enumerable extension of PA. With the same object types, indices $e, d : A \rightharpoonup B$ define the same arrow when

$$T \vdash \forall x \in A \left[(\varphi_e(x) \downarrow \leftrightarrow \varphi_d(x) \downarrow) \wedge (\varphi_e(x) \downarrow \Rightarrow \varphi_e(x) = \varphi_d(x)) \right]. \tag{4}$$

The product and sum types use the fixed effective numerical codings. The equality in (4) is a congruence: substitution into one uniform proof proves equality after precomposition, and the arithmetized composition equations prove equality after postcomposition. The structural identities and the parameter theorem have PA-provable versions, so they pass to the quotient.

The resulting category is a c^+-P-recursion category, with ranges, choice, and compatible distributive coproducts [5, §2, p. 646]. It is atomless and has insufficient constants. If T is Σ^0_1-sound, the nonzero connected domains are exactly the provable singleton domains; otherwise there are no nonzero connected domains. No S_T is locally connected [5, Theorem 4.3 and Corollary 4.4, pp. 657–658].

The pointwise constructions S' and S'_T use a different equivalence relation. Their proposed composition fails to be independent of representatives [10]; they are excluded from the catalogue. The uniform constructions S and S_T remain included.

2.3 Montagna–Sorbi's operator categories

We use the interleaving identification $\mathbb{B}^2 \cong \mathbb{B}$ from Section 1.1. An ordinary partial recursive operator can output a partial numerical function. A partial $\mathbb{B}$-operator is defined only when that output is total.

$\mathcal{B}$. This one-object category has carrier $\mathbb{B}$ and the extensional partial recursive $\mathbb{B}$-operators as arrows [4, Definition 2.1, p. 1027]. Its domains are exactly the lightface Π^0_2 subsets of Baire space. It does not have ranges for all arrows, and weak choice fails. It has atoms but not enough atoms [4, Propositions 2.1–2.3 and Lemma 2.2, pp. 1027–1030].

$\mathcal{B}_o$. The object is again Baire space. Arrows are all partial continuous maps with G_δ domains, with extensional equality [4, Lemma 2.5 and Definition 2.2, pp. 1030–1031]. This category contains $\mathcal{B}$, has enough atoms, and still lacks ranges and weak choice [4, Proposition 2.4].

Both categories are recursion categories [4, Proposition 3.1, p. 1032]. If (Φ_e) enumerates the partial recursive $\mathbb{B}$-operators, a common Turing morphism is

$$\tau(f \oplus g) = \Phi_{f(0)}(f^+ \oplus g).$$

A total compiler prefixes an operator index to the parameter function. The passage from $\mathcal{B}$ to $\mathcal{B}_o$ permits arbitrary real parameters and continuous compilers.

For subsets $D, E \subseteq \mathbb{B}$, Wadge reducibility $D \leq_W E$ means that $D = u^{-1}(E)$ for some total continuous $u : \mathbb{B} \to \mathbb{B}$. A G_δ set is G_δ-complete if every G_δ subset of Baire space reduces to it in this sense. Complete domains in $\mathcal{B}_o$ are the G_δ-complete sets under Wadge reducibility [4, Remark 3.1, p. 1034]. Some domains of $\mathcal{B}$ become complete in $\mathcal{B}_o$ without being complete in $\mathcal{B}$ [4, Corollary 3.4, pp. 1034–1035]. Thus the comparison changes the admitted reductions, a fact that must be recorded alongside any coalgebraic representation.

The atomless category $\mathcal{N}$. Let

$$X = \{f \in \omega^\omega : \text{every partial recursive } \varphi_e \subseteq f \text{ has finite domain}\}.$$

For $\Phi \in \mathcal{B}$, let Φ^X retain precisely those inputs in X whose outputs lie in X. The arrows of $\mathcal{N}$ form the category *generated* by these restricted operators. The assignment $\Phi \mapsto \Phi^X$ need not preserve composition. Montagna–Sorbi prove a normal form $P_1\Phi^X$ and then establish that $\mathcal{N}$ is a recursion category [4, Definitions 4.1–4.2, Lemma 4.3, Proposition 4.1, pp. 1038–1040]. It is atomless and has diagonal ranges. Its creativity notion relative to atoms does not imply completeness [4, Theorems 4.1–4.2].

Its carrier has a useful topological description.

Proposition 2.1. *The set X defining $\mathcal{N}$ is a dense G_δ subset of Baire space.*

Proof. For each e such that $\operatorname{dom}\varphi_e$ is infinite, the set C_e of total extensions of φ_e is closed. Given a basic cylinder, choose an argument in $\operatorname{dom}\varphi_e$ beyond its specified coordinates and extend the cylinder by a different value there. Hence C_e is nowhere dense. The complement of the union of these countably many C_e is X. The Baire category theorem proves density. □

This is a boldface G_δ description. Its indexing condition is not an effective enumeration of open complements.

2.4 Equivariant and existence constructions

Let G be a finite group and $\mathcal{C}$ a $+$-dominical recursion category with equality, in the terminology of Di Paola–Heller: coproducts have the required compatibility with the near product, and each diagonal has a range. In $\mathcal{C}^G$, write $G \otimes X$ for the free G-object, formed from the G-indexed coproduct of X. The full subcategories on free objects $G \otimes X$ and on objects $(G \otimes X) + X$, with trivial action on the last summand, are denoted $\mathcal{C}_1^G$ and $\mathcal{C}_0^G$. Both are recursion categories [2, Theorem 7.9.9, pp. 620–621]. These entries retain the finite group and the chosen base category.

Heller's existence theorem applies to uniformly generated isotypical $B^{\#}$-subcategories of total maps in locally connected iteration categories [6, Theorem 9.2]. His examples include set-based, cocommutative-coalgebra, functor-category, and frame-relative constructions [6, §10]. The generating data and iteration closure specify the resulting recursion category.

Lengyel's coalgebra construction uses partial morphisms of F-coalgebras as an ambient P-category [7]. The corrected relative existence theorem is as follows [9, Theorem 6.4]: let $\mathcal{C}, \mathcal{D}$ be iteration categories, with $\mathcal{D}$ separated, and let $L : \mathcal{C} \to \mathcal{D}$ be a faithful iteration functor. Every uniformly generated isotypical $B^{\#}$-subcategory $\mathcal{B}$ of the total maps of $\mathcal{C}$ has a recursion-category completion $\mathrm{Tur}(\mathcal{B})$. Here separated is the connected-domain separation condition called locally connected in Heller's terminology.

In ZFC, if $F : \mathbf{Set} \to \mathbf{Set}$ weakly preserves pullbacks and $\mathrm{Coalg}(F)$ has binary products, its partial maps with subcoalgebra domains form a relatively separated iteration category. The relative theorem therefore gives recursion categories from its uniformly generated isotypical $B^{\#}$-subcategories of total maps [9, Theorem 10.1]. These are construction schemes: an entry for an individual example includes the actual functor, isotypical object, generators, and required preservation hypotheses.

3 Representations and their 2-categorical comparisons

For an endofunctor $F : \mathbf{Set} \to \mathbf{Set}$, an F-coalgebra is a map $c : X \to FX$. A coalgebra morphism from (X, c) to (Y, d) is a function $u : X \to Y$ satisfying $du = (Fu)c$; these form $\mathrm{Coalg}(F)$. A subcoalgebra is a subset $D \subseteq X$ with a coalgebra structure making its inclusion a coalgebra morphism. A partial coalgebra morphism $(X, c) \rightharpoonup (Y, d)$ consists of a subcoalgebra domain $D \hookrightarrow X$ and a total coalgebra morphism $D \to Y$. Composition pulls back the second domain along the first map. We use this construction when subcoalgebra inclusions are stable under these pullbacks and it supplies the restriction, zero, and binary near-product structures specified below. Write $\mathcal{A}_F = \mathrm{Par}(\mathrm{Coalg}(F))$. For constant functors and the Moore functor used in Section 4, domains and composition have this form.

A representation is a faithful functor $J : \mathcal{R} \to \mathcal{A}_F$ that preserves restrictions and zeros and has canonical total isomorphisms

$$c^J_{X,Y} = \langle Jp_X, Jp_Y \rangle : J(X \boxtimes Y) \longrightarrow JX \boxtimes JY.$$

Thus preservation of products means preservation through their canonical comparison maps. Their associativity and symmetry compatibilities follow from the product equations. One may also require preservation of distributive binary coproducts, of ranges, or of both; the same requirements are imposed on every functor in a given comparison. Range preservation means $J(\mathrm{ran}\, f) = \mathrm{ran}(Jf)$ whenever these ranges are required. A Turing morphism remains universal for the represented arrows by applying J to its factorization equation.

For such functors $P, Q : \mathcal{C} \to \mathcal{D}$, a *structure-compatible natural transformation* $\theta : P \Rightarrow Q$ has total components $\theta_X : PX \to QX$, satisfies $Qf\,\theta_X = \theta_Y\,Pf$ for every $f : X \to Y$, and obeys

$$c^Q_{X,Y}\theta_{X \boxtimes Y} = (\theta_X \boxtimes \theta_Y)c^P_{X,Y}. \tag{5}$$

When coproduct preservation is required, put $d^P_{X,Y} = [Pi_X, Pi_Y] : PX + PY \to P(X+Y)$ and require $\theta_{X+Y} d^P_{X,Y} = d^Q_{X,Y}(\theta_X + \theta_Y)$. These equations use the canonical comparisons, so they are stable under vertical composition and whiskering.

To compare different catalogue entries as well as representations of one entry, use the following construction.

Definition 3.1. An object is a triple $(\mathcal{R}, F, J)$ as above. A 1-cell $(\mathcal{R}, F, J) \to (\mathcal{S}, G, J')$ is a triple (H, K, κ) consisting of structure-preserving functors $H : \mathcal{R} \to \mathcal{S}$ and $K : \mathcal{A}_F \to \mathcal{A}_G$, with an invertible structure-compatible natural transformation $\kappa : KJ \Rightarrow J'H$. A 2-cell to (H', K', κ') is a pair of structure-compatible natural transformations with total components, $\alpha : H \Rightarrow H'$ and $\eta : K \Rightarrow K'$, satisfying

$$(J'\alpha)\kappa = \kappa'(\eta J).$$

In particular, the component $\kappa_X : K(JX) \to J'(HX)$ is a total isomorphism, and its naturality equation is $J'(Hf)\kappa_X = \kappa_Y K(Jf)$ for $f : X \to Y$. The equation defining a 2-cell is an equality of transformations $KJ \Rightarrow J'H'$. The square being compared is

$$\begin{array}{ccc} \mathcal{R} & \xrightarrow{H} & \mathcal{S} \\ {\scriptstyle J}\downarrow & & \downarrow{\scriptstyle J'} \\ \mathcal{A}_F & \xrightarrow[K]{} & \mathcal{A}_G. \end{array}$$

Composition is pasting: for a second 1-cell (H_2, K_2, κ_2), the composite has functors H_2H, K_2K and comparison $(\kappa_2 H)(K_2\kappa)$. The 2-category laws follow from those of functors and natural transformations. Restricting to a fixed $\mathcal{R}$, with $H = \mathrm{id}_{\mathcal{R}}$ and $\alpha = \mathrm{id}$, recovers $\mathrm{Rep}(\mathcal{R})$ of [12].

An equivalence in $\mathrm{Rep}(\mathcal{R})$ includes an equivalence of the whole ambient targets. An equivalence of the categories of represented arrows is weaker. In fact, all faithful represented images of the same source, with source-object tags retained, are equivalent to that source. Classification of representations therefore requires the ambient category or additional effective and topological data.

3.1 Classification for constant targets

For a set E, coalgebras of the constant functor $F_E(X) = E$ are labelled sets $a : X \to E$. Their partial-map category $\mathbf{Pfn}_E$ has label-preserving partial functions as arrows. Products of total coalgebras are fibre products over E. A representation $J : \mathcal{R} \to \mathbf{Pfn}_E$ has fibre functors $J_e : \mathcal{R} \to \mathbf{Pfn}$ for $e \in E$.

Theorem 3.2. *Let $J : \mathcal{R} \to \mathbf{Pfn}_E$ and $J' : \mathcal{R} \to \mathbf{Pfn}_D$ be representations with the same specified preservation requirements. They are equivalent in* $\mathrm{Rep}(\mathcal{R})$ *if and only if there is a bijection $\sigma : E \to D$ and structure-compatible natural isomorphisms $J_e \cong J'_{\sigma(e)}$ for all $e \in E$.*

Proof. A restriction equivalence $K : \mathbf{Pfn}_E \simeq \mathbf{Pfn}_D$ restricts to an equivalence of total-map categories $\mathbf{Set}/E \simeq \mathbf{Set}/D$. It preserves the terminal object of these total-map categories and its subobject lattice. The latter is $\mathcal{P}(E)$ or $\mathcal{P}(D)$. Atoms are precisely singleton labels, so K induces a bijection $\sigma : E \to D$.

Every labelled set is the coproduct of its singleton labelled elements. Preservation of these coproducts identifies K, naturally on total maps, with relabelling along σ. A partial map factors as the partial inverse of its domain inclusion followed by a total map. Restriction preservation extends this identification to partial maps. Thus $KJ \cong J'$ gives exactly the displayed fibre isomorphisms. Conversely, relabelling along σ is a structured equivalence, and the fibre isomorphisms assemble to its required comparison with J'. The inverse relabelling gives the inverse 1-cell. □

The label set, including labels unused by the image, is therefore an ambient invariant. The theorem classifies the constant-target part of $\mathrm{Rep}(\mathcal{R})$; comparisons into larger coalgebra categories supply additional 1-cells.

4 Countable representations and integer-set coalgebras

4.1 A countable representation theorem

Proposition 4.1. *Every nontrivial prodominical P-category with countably many objects and arrows has a faithful representation in $\mathbf{Pfn}_E$ with E countable and every represented carrier countable. It preserves restrictions, zeros, and binary near products, and reflects totality. If distributive binary coproducts exist, such a representation can also preserve them.*

Proof. Split restriction idempotents: the objects of the resulting category $\mathcal{K}$ are (X, e) with $e = \overline{e}$, and arrows satisfy $f = dfe$. Its identity at (X, e) is e, and it is countable. Regard each original object X as $(X, 1_X)$. Use its nonzero objects A as labels. The A-fibre of J_0X is the set of total $h : A \to X$; define $J_0f(h) = fh$ precisely when fh is total.

A total composite has total inner factor, so this rule respects composition and definedness. The identity $\overline{fh} = \overline{\overline{f}h}$ proves restriction preservation; a zero composite cannot be total on nonzero A. Total maps into a restriction product are exactly pairs of total maps with the same source, proving product preservation. To distinguish f from g, split $\overline{f} = mp$. If $f \neq 0$, m is a generalized element on which J_0f is defined. Equality $J_0f = J_0g$ would give $fm = gm$, hence $f = g\overline{f}$. Reversing the roles gives $g = f\overline{g}$, and therefore $f = g$. The case $f = 0$ follows from the reverse argument. This proves faithfulness. Restriction preservation and faithfulness give reflection of totality. All fibres and labels are countable.

Suppose now that binary coproducts are distributive. The split category $\mathcal{K}$ inherits these coproducts and their distributivity. A *summand domain* on A is the restriction idempotent that is the identity on one side of a decomposition $A \cong A_0 + A_1$ and zero on the other. Write $B(A)$ for the set of these idempotents.

We first record the decomposition property used below. A total map $h : A \to X + Y$ decomposes A into the inverse images of the two summands. Indeed, put $q_0 = [1_X, 0]$, $q_1 = [0, 1_Y]$, and $e_i = \overline{q_i h}$. Distributivity gives the total decision map

$$d = (p_A + p_A)\delta^{-1}\langle 1_A, h\rangle : A \longrightarrow A + A,$$

where $\delta : (A \boxtimes X) + (A \boxtimes Y) \to A \boxtimes (X + Y)$ is the distributivity isomorphism. If $e_i = j_i p_i$ are splittings in $\mathcal{K}$, then $[j_0, j_1] : A_0 + A_1 \to A$ has inverse $(p_0 + p_1)d$. To check the inverse identities, let $k_i : A \to A + A$ be the injections, $r_0 = [1_A, 0]$, $r_1 = [0, 1_A]$, and $\nabla = [1_A, 1_A]$. The product equations and naturality of distributivity give

$$\nabla d = 1_A, \qquad r_i d = e_i, \qquad (h + h)d = (i_X + i_Y)h.$$

For the middle identity, for example, $r_0(p_A + p_A) = p_A(1_A \boxtimes q_0)\delta$; the two sides agree on each summand. For the last, use $(h \boxtimes 1)\langle 1_A, h\rangle = \Delta h$ and check $(p_{X+Y} + p_{X+Y})\delta^{-1}\Delta = i_X + i_Y$ on the two injections. Applying $q_0 + q_1$ gives $(q_0 h + q_1 h)d = h$. Since h is total, the restriction axioms give $(e_0 + e_1)d = d$. They also give $de_i = k_i e_i$ from $\overline{r_i} = k_i r_i$ and $r_i d = e_i$. Consequently

$$[j_0, j_1](p_0 + p_1)d = [e_0, e_1]d = \nabla(e_0 + e_1)d = 1_A.$$

The reverse composite is the identity on each summand because $dj_i = k_i j_i$ and $p_i j_i = 1_{A_i}$. This proves the decomposition used in the germ construction of [11, Appendix A]. Pulling one decomposition

back along the injections of another gives a common four-part decomposition. Consequently $B(A)$ is a Boolean algebra: meet is composition, complement exchanges the two summands, and join takes their union. A summand of a summand is again a summand.

Use every pair (A, U), where A is nonzero and U is an ultrafilter on $B(A)$, as a label. An arrow $h : A \to X$ is *admissible at* U if $e \leq \overline{h}$ for some $e \in U$. Two admissible arrows represent the same germ when

$$[h]_U = [k]_U \quad \Longleftrightarrow \quad \text{there is } e \in U \text{ with } e \leq \overline{h}\,\overline{k} \text{ and } he = ke.$$

Intersections of supporting members of U prove transitivity, so this is an equivalence relation. Let the (A, U)-fibre of JX be the set of these germs, and define the partial action by

$$Jf([h]_U) = [fh]_U \qquad \text{if } fh \text{ is admissible at } U.$$

If h and k agree on a member of U, intersect that member with one supporting fh. On the intersection, fh and fk are defined and equal. Thus both definedness and value are independent of the representative. Admissibility of gfh implies that of fh, which proves functoriality. The identity $\overline{f\overline{h}} = \overline{fh}$ proves preservation of restrictions, and a proper ultrafilter contains no zero member, so $J0$ is nowhere defined.

At each label, pairing gives a bijection

$$J(X \boxtimes Y)_{(A,U)} \longrightarrow JX_{(A,U)} \times JY_{(A,U)}, \qquad [h]_U \longmapsto ([p_X h]_U, [p_Y h]_U).$$

Its inverse pairs representatives after intersecting their supporting members. These bijections commute with postcomposition and preserve projections and diagonals. For a germ into $X + Y$, restrict a representative to a supporting summand $e \in U$ and apply the decomposition property. Its two inverse-image summands have disjoint domains whose join is e. Exactly one belongs to U, so the germ factors through exactly one coproduct injection. The two resulting images are disjoint: applying $[1_X, 0]$ to an equality between them would make a supported arrow zero. Hence $J(X + Y) \cong JX \amalg JY$, with the images of the injections as coproduct injections.

To prove faithfulness, suppose $Jf = Jg$ and split $\overline{f} = mp$, with $m : A \to X$ total. If $f \neq 0$, then A is nonzero and fm is total. At every ultrafilter U on $B(A)$, equality of the two represented partial functions supplies some $e \in U$ such that

$$e \leq \overline{gm}, \qquad fme = gme.$$

These agreement domains have a finite subfamily whose join is 1_A. Otherwise their complements would have the finite intersection property and would extend to an ultrafilter meeting none of them. Refine the finite cover to a disjoint summand partition. Its injections are jointly epimorphic, so $fm = gm$, and hence $f = g\overline{f}$. This equation also holds when $f = 0$. Interchanging f and g gives equality by antisymmetry of the restriction order. Thus J is faithful.

Each fibre is countable because its germs are represented by arrows of $\mathcal{K}$. For each unequal parallel pair f, g, choose one label at which Jf and Jg differ, and retain just these labels. There are countably many pairs. The restricted functor remains faithful, and all the preservation arguments above apply separately at each retained label. Its carriers are countable unions of countable fibres, as required. □

Consequently the classical, fixed-oracle, lightface arithmetical and analytical, syntactic, and recursive-operator entries with countable arrow sets have countable representations. This conclusion concerns cardinality. An effective presentation must additionally specify computable structure and represented arrow maps.

4.2 Binary observations of integer subsets

Take $M(Y) = 2 \times Y^{\{+,-\}}$. For $A \subseteq \mathbb{Z}$, let Q_A have carrier $\mathbb{Z}$, output χ_A, and transitions $n \mapsto n+1$ and $n \mapsto n-1$. Call A aperiodic if $A + k = A$ implies $k = 0$.

Proposition 4.2. *Every coalgebra morphism $Q_A \to Q_B$ is a translation $n \mapsto n + k$ with $B = A + k$. For aperiodic A, B, their translation classes therefore classify the Q_A up to isomorphism, and between two such coalgebras there is at most one morphism. Each Q_A has no proper nonempty subcoalgebra.*

Proof. Commutation with both transitions forces $h(n) = h(0) + n$ for every integer n. Preservation of output says $\chi_B(n + h(0)) = \chi_A(n)$, exactly the displayed translation condition. Two allowable translations differ by a period of A. A nonempty transition-closed subset of $\mathbb{Z}$ contains every integer. □

The same argument classifies all nonempty transitive binary systems with inverse transitions. Put $\mathrm{Stab}(A) = \{k \in \mathbb{Z} : A + k = A\}$.

Proposition 4.3. *A nonempty binary Moore coalgebra with mutually inverse transitions and no proper nonempty subcoalgebra is isomorphic to*

$$Q(H, A) = (\mathbb{Z}/H, o_A, \delta_+, \delta_-), \qquad H \leq \mathrm{Stab}(A),$$

where $A \subseteq \mathbb{Z}$ and $o_A : \mathbb{Z}/H \to 2$ is given by $o_A(n + H) = \chi_A(n)$, and the transitions add or subtract 1. The containment $H \leq \mathrm{Stab}(A)$ makes this output well defined. Two such coalgebras are isomorphic exactly when their subgroups H agree and their observation sets are integer translates. Moreover,

$$\mathrm{End}(Q(H, A)) = \mathrm{Aut}(Q(H, A)) \cong \mathrm{Stab}(A)/H.$$

Proof. The inverse transitions define a $\mathbb{Z}$-action. Its orbits are subcoalgebras, so the action is transitive. Choose a state q; its stabilizer is H, and let A consist of the integers n for which the output at $n \cdot q$ is 1. This identifies the carrier with $\mathbb{Z}/H$ and implies $H \leq \mathrm{Stab}(A)$. Changing the chosen state translates A and leaves H unchanged. An equivariant map between two transitive $\mathbb{Z}$-sets is determined by the image of the zero coset. For an isomorphism the stabilizers must agree; output preservation gives exactly the translation condition. For an endomorphism the allowable cosets are $\mathrm{Stab}(A)/H$, and every such translation is invertible. □

Thus the rigid systems in this class have $H = \mathrm{Stab}(A)$. They are the shift orbits of integer subsets, with finite cycles for periodic subsets and infinite orbits for aperiodic subsets.

For a countable label set choose $E \subseteq \omega$ and put $A_e = \{0, e+1\}$. These sets are aperiodic and pairwise nontranslate. For a labelled set (X, a) define $\Phi(X) = X \times \mathbb{Z}$, with

$$o(x, n) = \chi_{A_{a(x)}}(n), \qquad \delta_+(x, n) = (x, n+1), \quad \delta_-(x, n) = (x, n-1).$$

For a label-preserving partial function $f : D \to Y$, put $\Phi(f) = f \times \mathrm{id}_{\mathbb{Z}}$ with domain $D \times \mathbb{Z}$.

Theorem 4.4. *The functor $\Phi : \mathbf{Pfn}_E \to \mathrm{Par}(\mathrm{Coalg}(M))$ is fully faithful and preserves restrictions, zeros, coproducts, and all coalgebra products indexed by nonempty sets on total maps, hence binary near products. Composing it with the representation in Proposition 4.1 gives a faithful countable coalgebraic representation with binary observations.*

Proof. A subcoalgebra of $X \times \mathbb{Z}$ is $D \times \mathbb{Z}$ because both translations are available. A morphism on each copy of $\mathbb{Z}$ lands in one target copy; Proposition 4.2 and the chosen markers force preservation of its label and integer phase. It is therefore exactly $f \times \mathrm{id}$ for a labelled partial function f. This proves full faithfulness and the assertions about domains and zeros.

For a family $(X_i, a_i)_{i \in I}$ with $I \neq \varnothing$, put

$$\prod_E X_i = \{(x_i) \in \prod_{i \in I} X_i : a_i(x_i) \text{ is independent of } i\}.$$

Its label is this common value. The coalgebra product of the $\Phi(X_i)$ is $(\prod_E X_i) \times \mathbb{Z}$. To verify the universal property, take any cone of total coalgebra maps into the factors. At a state of its vertex, all cone components have the same output after every input word. In the image objects these outputs determine the marker label and integer phase, so both agree in every coordinate. The component maps thus pair uniquely into the stated object and commute with the transitions. Coproducts are disjoint unions with inherited structure. The product description also respects partial domains. □

The marker systems are uniformly computable in e. The generalized element or germ construction may nevertheless give noneffective represented arrows. For an effective refinement, computability of these partial actions is a separate mathematical requirement.

Corollary 4.5. *A nontrivial prodominical P-category with at most $2^{\aleph_0}$ objects and arrows has a faithful representation by coalgebras for $M(Y) = 2 \times Y^{\{+,-\}}$ whose states can be coded by subsets of ω. Restrictions, zeros, and binary near products are preserved; distributive binary coproducts can also be preserved when present.*

Proof. The proof of Proposition 4.1, with an infinite cardinal κ in place of countability, bounds labels and carriers by κ. In the germ construction there are at most κ parallel arrow pairs to separate, and every fibre has size at most κ. Apply this with $\kappa = 2^{\aleph_0}$. For the marker construction choose distinct subsets $A_e \subseteq \omega$ containing 0. Their minimum is 0, so each is aperiodic as a subset of $\mathbb{Z}$, and distinct such sets cannot be translates. There are $2^{\aleph_0}$ choices. The proof of Theorem 4.4 applies to this family. Each resulting carrier has cardinal at most $2^{\aleph_0}$ and therefore injects into $\mathcal{P}(\omega)$; transport its coalgebra structure along the injection. □

This applies to $\mathcal{B}_o$ as well as the countable-arrow examples. There are at most continuum many G_δ domains, and a continuous map from each such domain into Baire space is determined by its values on a countable dense subset. The resulting set-theoretic representation leaves the topological preservation problem in Section 6 to be solved.

5 Descriptive set theory and computation

5.1 Sets as behaviors and as codes

There are several useful ways for sets of integers to enter a representation. A state may be an oracle, a function coded by its graph, or a complete coalgebra presentation coded as a real. Alternatively, an integer set may be the observable behavior of a state. These interpretations impose different conditions on morphisms.

The graph coding of Section 1.1 identifies the carriers of $\mathcal{B}$, $\mathcal{B}_o$, and $\mathcal{N}$ with subspaces of $\mathcal{G} \subseteq 2^\omega$. It also makes their operators partial maps between sets of graph codes, by conjugating with $f \mapsto \chi_f$. To obtain a coalgebraic representation one must choose observations and transitions for which the admitted arrows are coalgebra morphisms.

5.2 Binary and positive observations

The coalgebra

$$\mathcal{P}(\omega) \longrightarrow 2 \times \mathcal{P}(\omega), \qquad S \longmapsto (\chi_S(0), \{n : n+1 \in S\})$$

is final for $2 \times -$ [13, Corollary 2.3.6]. For $c = (o, \delta) : X \to 2 \times X$, its unique behavior map is $b(x) = \{n : o(\delta^n x) = 1\}$. Every S has a countable realization on ω, with successor transition and output χ_S. A realization with computable initial state and total computable binary output and transition exists exactly when S is recursive: iteration computes each membership bit, and the successor construction proves the converse. A finite-state realization exists exactly when the characteristic sequence is ultimately periodic.

Positive information uses Sierpiński space $\mathbb{S}$, whose open sets are $\varnothing, \{1\}, \{0, 1\}$. The Scott topology on $\mathcal{P}(\omega)$ has basic opens $\{S : D \subseteq S\}$ for finite $D \subseteq \omega$, and identifies this space with $\mathbb{S}^\omega$. The same head and tail make it final for $\mathbb{S} \times -$ in topological spaces: the displayed behavior map is continuous because each positive membership test has open inverse image. More explicitly, an enumeration name for $S \subseteq \omega$ is a sequence $p \in \omega^\omega$ satisfying

$$S = \{n : (\exists k)\ p(k) = n + 1\}.$$

The value 0 supplies a blank entry, and repetitions are allowed. The set has a computable enumeration name exactly when it is c.e. By contrast, the characteristic-function representation names S by its binary sequence χ_S; that name is computable exactly when S is recursive [17, 18].

Enumeration operators supply another direct connection. A relation $W \subseteq \mathrm{Fin}(\omega) \times \omega$ defines

$$\Psi_W(S) = \{n : (D, n) \in W \text{ for some finite } D \subseteq S\}.$$

These are Scott-continuous maps; conversely, a Scott-continuous map is determined by its values on finite sets and has such a form. For the converse, if $n \in \Psi(S)$, Scott openness of the inverse image of $\{T : n \in T\}$ supplies a finite $D \subseteq S$ with $n \in \Psi(D)$. Thus one can take $W = \{(D, n) : n \in \Psi(D)\}$. To make effectiveness precise, code a finite set D by $\ulcorner D \urcorner = \sum_{k \in D} 2^k$, and code its rule for n by $\langle \ulcorner D \urcorner, n \rangle$. If this set of rule codes is c.e., Ψ_W is an enumeration operator: while enumerating S, output n whenever all members of some enumerated rule's finite premise D have appeared. The information supplied by observations thus distinguishes decision from semidecision.

5.3 Why observable behavior alone is insufficient

Proposition 5.1. *Let $\mathcal{R}$ be a nontrivial recursion category and $J : \mathcal{R} \to \mathrm{Par}(\mathrm{Coalg}(F))$ a faithful representation preserving totality. Suppose F has a final coalgebra B. For every object X of $\mathcal{R}$, the behavior morphism $JX \to B$ is not monic in* $\mathrm{Coalg}(F)$.

Proof. An object Q with monic map to a final object is subterminal: for any $u, v : P \to Q$, their composites into the final object agree, so $u = v$. Choose a Turing morphism $\tau : W \boxtimes U \to V$. The isotype supplies a total isomorphism $h : W \boxtimes U \to V$. It is nonzero since every object of a nontrivial isotype is nonzero. Apply the Turing property to h and zero. It supplies distinct total endomorphisms $g_h, g_0 : W \to W$, since equal parameter maps would give $h = 0$. Their faithful images are distinct total coalgebra endomorphisms of JW. Hence JW is not subterminal. Every JX is isomorphic to JW, which proves the claim. □

The copies in Theorem 4.4 retain exactly this necessary kind of multiplicity: two states can have the same label and phase but different copy indices. Forgetting those indices can identify distinct represented arrows. Domains are also relevant: a partial map descends along a behavioral quotient only when its domain is saturated under behavioral equivalence [12, Proposition 3.3].

5.4 Equivalence relations on coalgebra codes

Fix a finite or countable input alphabet A, with a chosen coding of its finite words. The space of pointed binary Moore coalgebras on carrier ω is

$$\mathcal{C}_A = 2^\omega \times (\omega^\omega)^A \times \omega.$$

A code $(o, (\delta_a)_{a\in A}, x)$ specifies the output function $o : \omega \to 2$, one total transition $\delta_a : \omega \to \omega$ for each a, and an initial state $x \in \omega$. All coordinates have their product topologies, with ω discrete. The behavior of this pointed system is

$$L_x = \{w \in A^* : o(\delta_w x) = 1\} \in \mathcal{P}(A^*).$$

For $w = a_1 \cdots a_k$, set $\delta_w = \delta_{a_k} \cdots \delta_{a_1}$, so the letters are read from left to right; $\delta_\varepsilon = 1$ for the empty word. The map $\beta : \mathcal{C}_A \to 2^{A^*}$ sending a code to χ_{L_x} is continuous. To determine one output coordinate w, one reads only the initial state and the finitely many table entries encountered while following w.

A bisimulation between two such systems is a relation on their carriers whose related states have equal outputs and whose a-successors are related for every $a \in A$. Related states have equal languages, by induction on word length. Conversely, the relation of having equal languages is a bisimulation, since equal languages have equal empty-word outputs and equal residual languages after each letter. Hence pointed bisimilarity is

$$E_A = (\beta \times \beta)^{-1}(\Delta_{2^{A^*}}).$$

It is closed because β is continuous and the diagonal of 2^{A^*} is closed. An equivalence relation on a standard Borel space is *smooth* if a Borel map into 2^ω reduces it to equality. Coding A^* by integers makes β such a reduction, so E_A is smooth. For finite A, the behaviors realized by finite-state systems are exactly the regular languages [14].

For countable nondeterministic labelled transition systems, Sánchez Terraf proves that bisimilarity is $\mathbf{\Sigma}^1_1$-complete [19]. Thus the chosen transition functor changes the descriptive complexity of behavioral classification. Isomorphism of presentations and equivalence of representations are separate equivalence relations.

5.5 Moore coalgebras and higher-type operators

The finite-word behavior of a pointed Moore coalgebra also determines an operator on infinite input streams. Let A be a nonempty countable alphabet. For a stream $\alpha \in A^\omega$, write $\alpha \restriction n = \alpha(0) \cdots \alpha(n-1)$ for its prefix of length n, including the empty prefix when $n = 0$. A pointed Moore coalgebra with output $o : X \to 2$, transitions $\delta_a : X \to X$, and initial state $x \in X$ induces

$$T_x : A^\omega \longrightarrow 2^\omega, \qquad T_x(\alpha)(n) = o(\delta_{\alpha\restriction n} x). \tag{6}$$

The zeroth output is $o(x)$, emitted before the first input is read. We call an operator $T : A^\omega \to 2^\omega$ *strictly causal* if, for all $\alpha, \gamma \in A^\omega$ and $n \in \omega$,

$$\alpha \restriction n = \gamma \restriction n \quad \Longrightarrow \quad T(\alpha)(n) = T(\gamma)(n). \tag{7}$$

Thus the output at time n depends only on the preceding n inputs. Every T_x is strictly causal, and every strictly causal operator is continuous for the product topologies: any finite set of output coordinates is determined by a sufficiently long finite input prefix.

When $A = \omega$, the displayed operator has type $(\omega \to \omega) \to (\omega \to 2)$. With numerical and Boolean values at type level zero and streams at level one, this is an operator of type level two. Each coordinate $\alpha \mapsto T_x(\alpha)(n)$ is a Boolean-valued functional of type two. The coalgebra's output and transition maps give a state-based presentation of this operator.

For the effective statements, take $A = \omega$ or $A = \{0, \dots, k-1\}$ with $k \geq 1$. An operator $T : A^\omega \to 2^\omega$ is *Type-2 computable* if one oracle algorithm, given n and oracle access to α, halts with output $T(\alpha)(n)$ for every α and n. Each such computation reads finitely many coordinates of its oracle. A *computable Moore presentation* has carrier ω, a computable output $o : \omega \to 2$, a jointly computable transition map $(a, s) \mapsto \delta_a(s)$ from $A \times \omega$ to ω, and a specified initial state in ω.

Proposition 5.2. *For every nonempty countable alphabet A, the strictly causal operators $A^\omega \to 2^\omega$ are exactly the operators induced by pointed Moore coalgebras with countable state sets. If $A = \omega$ or $A = \{0, \dots, k-1\}$ with $k \geq 1$, a strictly causal operator is Type-2 computable if and only if it has a computable Moore presentation.*

Proof. Strict causality of a Moore operator follows from (6). Conversely, let T be strictly causal and choose $a_0 \in A$. Use the countable state set A^*, initial state ε, and transitions and outputs

$$\delta_a(s) = sa, \qquad o(s) = T(sa_0^\omega)(|s|).$$

Here sa_0^ω is the infinite stream beginning with s and continuing with a_0 forever. Strict causality makes this output independent of the chosen tail. After reading $\alpha \restriction n$, the state is exactly that prefix, and its output is $T(\alpha)(n)$.

A computable Moore presentation computes $T_x(\alpha)(n)$ by reading the first n inputs, applying the corresponding transitions, and evaluating the output. For the converse, give A^* a computable bijection with ω and use the preceding prefix construction. Such a bijection is obtained by length-lexicographic enumeration for finite A; for $A = \omega$, enumerate words by increasing $|s| + \sum_{i<|s|} s(i)$, with lexicographic order within each finite group. Concatenation is computable in either coding. If T is Type-2 computable, the value $T(sa_0^\omega)(|s|)$ is computable uniformly in s, since its oracle stream is uniformly computable from s. Thus the output and joint transition maps are computable. □

The interpretation preserves exactly the behavioral equality of §5.4. For two pointed systems with languages L_x and L_y,

$$T_x = T_y \quad \Longleftrightarrow \quad L_x = L_y. \tag{8}$$

Equality of languages gives equality of every output in (6). Conversely, extend any word w to wa_0^ω and inspect output $|w|$ to recover its membership in the language. Hence the equivalence relation E_A also classifies these pointed systems by equality of their induced stream operators.

The code-to-behavior map $\beta : \mathcal{C}_A \to 2^{A^*}$ has a uniform effective interpretation as well. For the effective alphabets above, name a code by its initial state and oracle access to the tables $o(s)$ and $\delta_a(s)$; the pairing function (1) combines the table indices into numerical coordinates. Given a word w, an oracle algorithm follows its finitely many transitions and queries the final output. It therefore computes $\beta(c)(w)$ uniformly from the name of c, even when the tables themselves are noncomputable. With these representations, β is a Type-2 computable map between spaces of codes.

The timing convention distinguishes Moore operators from the full class of continuous stream operators. For example, the left shift $S : 2^\omega \to 2^\omega$, defined by $S(\alpha)(n) = \alpha(n+1)$, is computable and continuous but fails (7). An operator is *causal* if its output at time n depends only on the input prefix of length $n+1$. Deleting the zeroth output of a Moore operator gives a causal operator. Mealy coalgebras for $X \mapsto (B \times X)^A$ produce an output with each input; their final coalgebra consists of all causal functions $A^\omega \to B^\omega$ [15, Section 2.2, Theorem 2.3].

General continuous stream operators can read a variable finite number of inputs before emitting each output. Garner studies such stream processors and gives an extensional coalgebraic characterization of the continuous functions $A^\omega \to B^\omega$ [16, Theorem 1.1]. For $A = B = \omega$, these are operators on Baire space. The total Type-2 computable operators and total continuous operators are, respectively, the total arrows of Montagna–Sorbi's $\mathcal{B}$ and $\mathcal{B}_o$. Extending these presentations to their partial arrows requires accounting for the Π^0_2 and G_δ domains described in §2.3. This is part of the first classification problem below.

6 Specific classification problems

The catalogue suggests three concrete comparisons.

1. For $\mathcal{B} \subseteq \mathcal{B}_o$, construct compatible coalgebraic representations that preserve the distinction between effective and continuous reductions. Determine which representations recover the Π^0_2 and G_δ domain structures and the completeness change identified by Montagna–Sorbi.

2. For S_T, compare the countable generalized-element and germ representations as T varies. Determine which coalgebraic invariants detect provable equality, ranges, and the connected-domain distinction of Di Paola–Montagna's Theorem 4.3.

3. For analytical and admissible entries, identify the uniformization and coding needed for effective coalgebraic representations. Compare the resulting 1-cells with numerical oracle inclusions and with pointclass-preserving transformations.

For the shift coalgebras of Proposition 4.2, the first isomorphism classification is explicit: integer translation of the observation set. The next problem is to classify representations assembled from these systems, retaining multiplicities, admissible domains, and the represented arrow family. Theorem 3.2 provides the constant-target comparison with which to begin.

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